P1 Wave packet

$$\vec{E}(x,0) = \vec{E}_0\, e^{i k_0 x}\, e^{-\frac{x^2}{2\ell^2}}$$

$$\vec{E}(\vec{k},0) = \int d^{\cancel{3}}\cancel{x}\; \vec{E}(x,0)\, e^{-ikx} = \vec{E}_0 \int_{-\infty}^{\infty} dx\; e^{i(k_0-k)x}\, e^{-\frac{x^2}{2\ell^2}}$$

$$= (x=\ell y) = \vec{E}_0\, \ell \int dy\; e^{i(k_0-k)\ell y}\, e^{-\frac{1}{2}y^2}$$

$$-\frac{1}{2}y^2 + i(k_0-k)\ell y = -\frac{1}{2}\left\{\left(y - i(k_0-k)\ell\right)^2 + (k_0-k)^2\ell^2\right\}$$

$$\vec{E}(k,0) = \vec{E}_0\, \ell \underbrace{\int_{-\infty}^{\infty} dy\; e^{-\frac{1}{2}(y+i(k-k_0)\ell)^2}}_{\sqrt{2\pi}}\; e^{-\frac{1}{2}(k-k_0)^2\ell^2}$$

$$= \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{-\frac{1}{2}(k-k_0)^2\ell^2}$$

Gaussian distribution in $\vec{k}$-space!

$$\vec{E}(k,t) = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{-\frac{1}{2}(k-k_0)^2\ell^2}\, e^{-i\omega(k)t}$$

$$\vec{E}(x,t) = \int_{-\infty}^{\infty} \frac{dk}{2\pi}\; \vec{E}(k,t)\, e^{ikx} = \sqrt{2\pi}\,\ell\, \vec{E}_0 \int \frac{dk}{2\pi}\; e^{-\frac{1}{2}(k-k_0)^2\ell^2} \cdot$$

$$\cdot\; e^{-i\omega(k)t} \cdot e^{i(k-k_0)x} \cdot e^{ik_0x}$$

$$(k=k_0+p) \quad = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{ik_0x} \int_{-\infty}^{\infty} \frac{dp}{2\pi}\; e^{-\frac{1}{2}p^2\ell^2}\, e^{+ipx}\, e^{-i\omega(k_0+p)t}$$

let us expand $\omega(p+k_0) \approx \omega(k_0) + \omega'(k_0)\,p + \frac{1}{2}\omega''(k_0)\,p^2$

$$= \omega_0 + \omega_0'\,p + \frac{1}{2}\omega_0''\,p^2$$

$$\Rightarrow \vec{E}(x,t) = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{ik_0x}\, e^{-i\omega_0 t} \int_{-\infty}^{\infty} \frac{dp}{2\pi}\; e^{ipx - \frac{1}{2}p^2\ell^2} \cdot$$

$$\cdot\; e^{-i\left(\omega_0' p + \frac{1}{2}\omega_0'' p^2\right)t}$$

We need to evaluate

$$J = \int_{-\infty}^{\infty} \frac{dp}{2\pi} \; e^{ip\underbrace{(x - \omega_0' t)}_{\beta}} \; e^{-\frac{1}{2}p^2 \underbrace{(\ell^2 + i\omega_0'' t)}_{\alpha}}$$

$$-\frac{1}{2}\alpha p^2 + i\beta p = -\frac{1}{2}\alpha\left(p^2 - 2\frac{i\beta}{\alpha}p\right) =$$

$$= -\frac{1}{2}\alpha\left\{\left(p - \frac{i\beta}{\alpha}\right)^2 + \frac{\beta^2}{\alpha^2}\right\} = -\frac{1}{2}\alpha\left(p - \frac{i\beta}{\alpha}\right)^2 - \frac{\beta^2}{2\alpha}$$

$$\Rightarrow \; J = e^{-\frac{\beta^2}{2\alpha}} \int_{-\infty}^{\infty} \frac{dp}{2\pi} \; e^{-\frac{\alpha}{2}\left(p - \frac{i\beta}{\alpha}\right)^2} =$$

$$= e^{-\frac{\beta^2}{2\alpha}} \frac{\sqrt{2\pi}}{\sqrt{\alpha}} \frac{1}{2\pi}$$

$$\vec{E}(x,t) = \frac{\ell \vec{E}_0 \, e^{i(k_0 x - \omega_0 t)}}{\sqrt{\ell^2 + i\omega_0'' t}} \exp\left\{-\frac{1}{2}\cdot\frac{(x - \omega_0' t)^2}{(\ell^2 + i\omega_0'' t)}\right\}$$

$$= \vec{E}_0 \, e^{i(k_0 x - \omega_0 t)} \frac{\ell}{\sqrt{\ell^2 + i\omega_0'' t}} \exp\left\{-\frac{(x - \omega_0' t)^2 (\ell^2 - i\omega_0'' t)}{\ell^4 + (\omega_0'' t)^2}\right\}$$

The packet moves with $v_g = \omega_0' = \left.\frac{d\omega}{dk}\right|_{k_0}$ = group velocity.
Its width increases as

$$\ell(t) = \sqrt{\ell^2 + \left(\frac{\omega_0'' t}{\ell}\right)^2}$$

therefore it becomes broader with rate $\frac{\omega_0'' t}{\ell}$.

Problem 2

$$E_x = A e^{ikx-i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2-\omega^2/c^2}\right), & z>0 \\ \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases} \qquad (1)$$

Similarly, E_z is given in the class notes:

$$E_z = ikAe^{ikx-i\omega t} \begin{cases} \dfrac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}, & z>0 \\ -\dfrac{\exp\left(z\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}\right)}{\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}}, & z<0 \end{cases} \qquad (2)$$

Magnetic field is given by

$$\vec{B} = -\frac{i}{\omega}\nabla\times\vec{E}$$

$$B_x = -\frac{i}{\omega}\left(\frac{\partial}{\partial y}E_z - \frac{\partial E_y}{\partial z}\right) = 0$$

(since $E_y=0$ and E_x, E_z are independent of y)

$$B_z = -\frac{i}{\omega}\left(\frac{\partial}{\partial x}E_y - \frac{\partial E_x}{\partial y}\right) = 0$$

$$B_y = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - ikE_z\right) \qquad (3)$$

We now substitute Eqs. (1) and (2) into Eq. (3):

for $z>0$:

$$B_y = -\frac{i}{\omega}\left(-\sqrt{k^2-\omega^2/c^2}\,Ae^{ikx-i\omega t}\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right) + k^2Ae^{ikx-i\omega t}\frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}\right) = \qquad (4)$$

$$= -\frac{i\omega}{c^2} A e^{ikx-i\omega t} \frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}$$

for $z<0$:

$$B_y = -\frac{i}{\omega}\left(\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, A e^{ikx-i\omega t} \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)\right.$$

$$\left. - k^2 A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}\right) \qquad (5)$$

$$= \frac{i\varepsilon\omega}{\varepsilon_0 c^2} A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}$$

By virtue of the dispersion equation for the surface plasmons:

$$\frac{\varepsilon}{\varepsilon_0} = -\frac{\sqrt{k^2-\frac{\omega^2\varepsilon}{c^2\varepsilon_0}}}{\sqrt{k^2-\frac{\omega^2}{c^2}}}$$

the two expressions (4) and (5) match at $z=0$.

Thus, $\vec{B}$ is indeed continuous at the surface.

Problem 3. The equation for the drift velocities are

$$\frac{d\vec{v}_1}{dt} = -\frac{e}{m_1}\vec{E} - \frac{e}{m_1}\vec{v}_1 \times \vec{B} - \frac{\vec{v}_1}{\tau} = 0,$$

$$\frac{d\vec{v}_2}{dt} = \frac{e}{m_2}\vec{E} + \frac{e}{m_2}\vec{v}_2 \times \vec{B} - \frac{\vec{v}_2}{\tau} = 0,$$

each equation is solved independently, similar to how it was done in class, then $\vec{j} = -en_1\vec{v}_1 + en_2\vec{v}_2$ The total conductivity is the sum of the two conductivities

$$\sigma_{ik} = \sigma_{ik}^{(1)} + \sigma_{ik}^{(2)}$$

where $$\sigma_{ik}^{(1)} = \frac{e^2 n_1 \tau_1}{(\omega_{c1}^2\tau_1^2 + 1)m_1}\begin{pmatrix} 1 & -\omega_{c1}\tau \\ \omega_{c1}\tau & 1 \end{pmatrix}$$

where $\omega_{c1} = \frac{eB}{m_1} > 0$

Similarly,

$$\sigma_{ik}^{(2)} = \frac{e^2 n_2 \tau_2}{(\omega_{c2}^2 \tau^2 + 1) m_2} \begin{pmatrix} 1 & \omega_{c2}\tau \\ -\omega_{c2}\tau & 1 \end{pmatrix} \qquad \omega_{c2} = \frac{eB}{m_2} > 0$$

The Hall current (perpendicular to both $\vec{E}$ and $\vec{B}$) is given by off-diagonal terms

$$j_H = \frac{-e^2 n_1 \tau_1^2 E \omega_{c1}}{(\omega_{c1}^2 \tau^2 + 1) m_1} + \frac{e^2 n_2 \tau_2^2 \omega_{c2} E}{(\omega_{c2}^2 \tau^2 + 1) m_2}$$

For strong magnetic field, such that

$$\omega_{c1}\tau \gg 1 \quad \text{and} \quad \omega_{c2}\tau \gg 1$$

the Hall current simplifies to

$$j_H = -\frac{e^2 n_1}{m_1 \omega_{c1}} E + \frac{e^2 n_2}{m_2 \omega_{c2}} = \frac{e(n_2 - n_1)E}{B}$$

We see that for equal number of electrons and holes (compensated semiconductor) the Hall effect vanishes. Note that for n-type and p-type semiconductors the Hall current has <u>opposite</u> sign.